Sulav Sapkota

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Education

Washington and Lee University — Expected May 2029
Intended Majors: Mathematics (B.S.) and Computer Science (B.S.) — **GPA:** 3.93/4.00 — *Lexington, VA*

- **Honors:** President's List 2025–2026
- **Pre-College Honors:** Gold Medal, National Chemistry Olympiad 2023; National Dynamite, National Olympiad of Informatics 2022; Gold Medal, Hydrocon 2.0 2022 (built a maze-solving robot using Arduino)
- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Intro to CS, Data Structures and Algorithms

Technical Skills

Languages & Tools: Python, C++20, R, SQL, JavaScript, React, Node.js, Django, Git/GitHub, Make, Unit Testing
ML and Geospatial: TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, geopandas, rasterio, QGIS

Experience

Washington and Lee University — June 2026 – Present
Machine Learning Research Assistant — *Lexington, VA*

- Built an end-to-end Python pipeline to predict the protected side of aerial fire-retardant drops from burn maps
- Processed 21,000+ drop polygons from 424 US wildfires with geopandas and rasterio in a reproducible workflow
- Engineered raster-split, unburned-distance and expanded-polygon features, boosting accuracy from 81% to 91%
- Evaluated Random Forest and XGBoost with fire-grouped 5-fold CV to avoid leakage, reaching 91.5% accuracy

Kathmandu Institute of Applied Science — June 2023 – June 2025
Machine Learning Researcher — *Lalitpur, Nepal*

- Built U-Net/VGG16 models to detect microplastics (82% binary test accuracy, 783 images); 5-class segmentation
- Added YOLOv8 detection (0.40 mAP50) and OpenCV post-processing to count particles against ground truth
- Won Best Poster at MoChWo 2024, an international conference in Kathmandu, presenting to 100+ researchers

Water Stories of Patan — June 2024 – August 2024
Student Researcher — *Lalitpur, Nepal*

- Managed a 6-person team on digital heritage project documenting the traditional stone water spouts (hitis) of Patan
- Surveyed 30+ locals across 10+ neighborhoods and interviewed NGOs, INGOs and agencies using snowball sampling
- Mapped water policy (1963–2015), ran a middle-school conservation workshop, and published findings on a website

Hello World I.T Solutions — January 2021 – June 2024
Software Developer — *Kathmandu, Nepal*

- Built the Django REST API (7 endpoints) and React frontend (fixtures, news, shop with cart) for a soccer club
- Built a live React/Node.js QI Dashboard with Chart.js: 12 health KPIs, 7 provinces, and a zoomable district map
- Built the React frontend for a team's AI-assisted mental health patient-tracking web portal, presented to UNICEF

Projects

Rumor Mill | *C++20, Python, pandas, matplotlib*

- Built a C++20 discrete-event gossip simulator with a priority-queue event engine, seeded RNG, loss and latency
- Wrote 6 unit tests (graph connectivity, time limit, same-seed reproducibility, loss extremes) and a Makefile build
- Ran 180 simulations on 1000 nodes; fanout 1 to 6 cut time to 99% coverage from 10.6 to 1.8 simulated time units
- Found total messages sent stayed ∼10–11k across fanouts, with 85–90% duplicates: fanout buys speed, not lower cost

Himalayan Quake Atlas | *Python, SQLite, pandas, Chart.js, Leaflet*

- Built a Python and SQLite pipeline that fetches, cleans and stores 10,125 USGS earthquakes from 1990 to 2026
- Wrote 16 pytest unit tests covering data cleaning, the database, and b-value recovery on synthetic catalogs
- Estimated the Gutenberg-Richter b-value (1.25 ± 0.03) and showed it swings from 0.79 to 1.27 with the cutoff
- Shipped an interactive Chart.js/Leaflet dashboard; regional b-values showed no statistically significant differences

Leadership / Extracurricular

Williams Investment Society — January 2026 – Present
Shadow — *Lexington, VA*

- Selected as 1 of 10 first-year members of the WIS shadow program, a student club managing a $30M+ portfolio
- Pitched GOOGL and VISA holdings using DCF, comparable company analysis, and qualitative sector research